OmniLabel AI™

Medical Device Label Change Intelligence Platform

Finds the label your change could miss

From Engineering Change Order (ECO) to Compliant Release

FDA 21 CFR Part 11 (Design Aligned) | EU MDR 2017/745 (Design Aligned) | ISO 15223-1 Label Symbols (Deterministic Checks)

1 The Problem
Silent Scope Omission

Engineering Change Orders (ECO)
Extend label life (e.g., 24 → 36 months) requires updating every affected label.

Broken PLM/ERP Links
Legacy systems (SAP, Teamcenter, Windchill) have missing or severed relationships.

Regulatory Risk
Outdated labels can trigger FDA 483 warnings, Class II recalls, and hospital quarantines.

2 Our Approach
AI Extracts. Rules Decide. Humans Sign.

Probabilistic AI Extracts
NLP + graph analysis to find potentially affected variants.

Deterministic Rules Decide
GS1 GTIN-14, ISO 15223-1 symbols, mathematical validation.

Qualified Humans Sign
Safe Scope Gate holds release until 100% of candidates are triaged and approved.

3 Dual Discovery & Safe Scope Gate
AI finds what you didn't link. Rules validate. Humans decide.

ECO Intake & Provenance → Dual Discovery (Forward + Inverse) → Safe Scope Gate (Human Triage) → Controlled Artwork Drafting



4 Deterministic Verification
Mathematical rules and computer vision inspection.

GS1 GTIN-14 Mod-10 Check
Validates device identifier syntax and check digit.

Inverse Heuristic Scoring
Combines family, sterility, shelf life and similarity (Jaccard).

OpenCV Computer Vision
Pixel-level comparison and contour analysis.

5 Audit & Human Approval
Compliant, traceable and tamper-evident.

SHA-256 Audit Chain
Immutable record of all decisions.

Qualified Reviewer
Human approval via Safe Scope Gate.

Timestamp & e-Signatures
21 CFR Part 11 compliant audit trail.

6 Test Results & Impact (Project-Reported)
Validates accuracy and compliance. Reduces business risk.

43 / 43 Tests Passing (100% Pass Rate)	**3.44 s** Execution Time (10 specialized modules)	**0** False Criticals (0%)	**100%** Audit Chain Integrity (28 / 28 records)

Regulatory Alignment (Design Aligned)

- FDA 21 CFR Part 11 — Electronic signatures, audit trails
- EU MDR 2017/745 — Labeling & UDI requirements
- ISO 15223-1 — Medical device label symbols